"Strings - Continuation"

* String Methods

Constructors
- new String ()
- new String (String)
- new String (char[])
- new String (byte[])
- new String (StringBuilder)
- new String (StringBuffer)

* String methods

length / Emptiness
- length ()
- isEmpty ()
- isBlank ()

Searching
- contains()
- indexOf()
- lastIndexOf()
- startsWith()
- endsWith()

Conversion
- valueOf()
- getBytes()

Advance
- intern()
- format()

character Accer
- charAt (int)
- toCharArray()

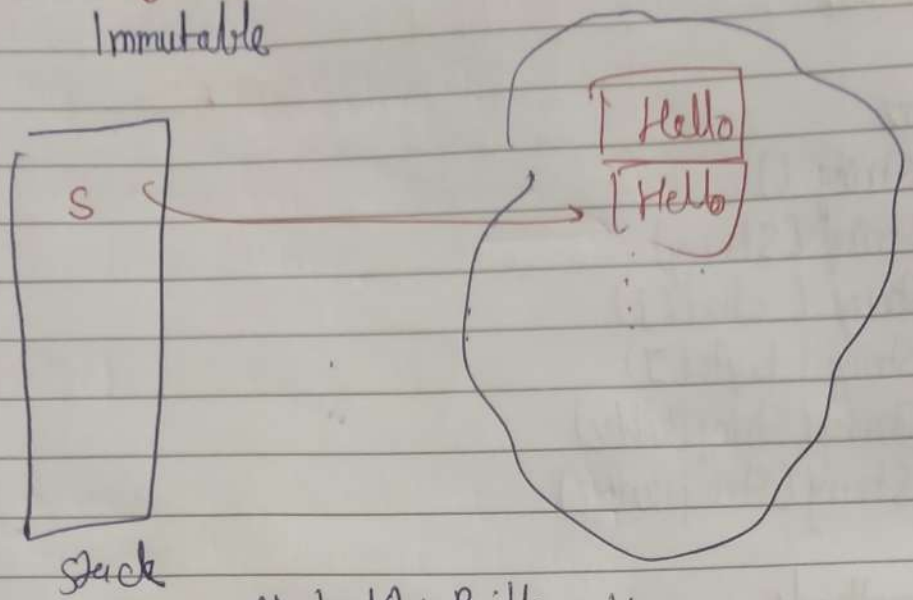
Comparison
- equals()
- equalsIgnoreCase()
- CompareTo()

Extraction / Transformation
- substring ()
- toUpperCase()
- toLowerCase()
- trim()
- strip()
- repeat()
- replace()
- replaceAll()
- split()
- join()

Strings → Method

↓

Immutable



AbstractStringBuilder

StringBuilder — mutable — StringBuffer

└ Not threadsafe (x)

└ Thread safe (✓)

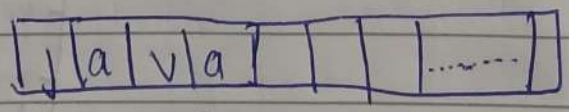
java.lang

```
class StringBuilder extends AbstractStringBuilder {
    byte [] values; //code
    int count;
}
```

initial capacity = 16

count = 4

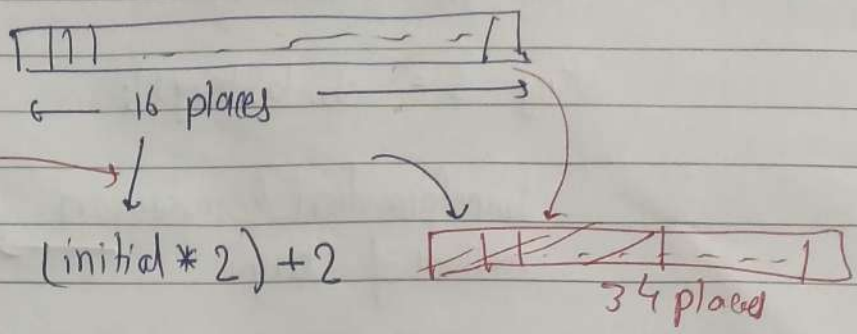
```
StringBuilder sb = new StringBuilder("java");
sb.append("is cool"); //count = 12
```



java. is. cool

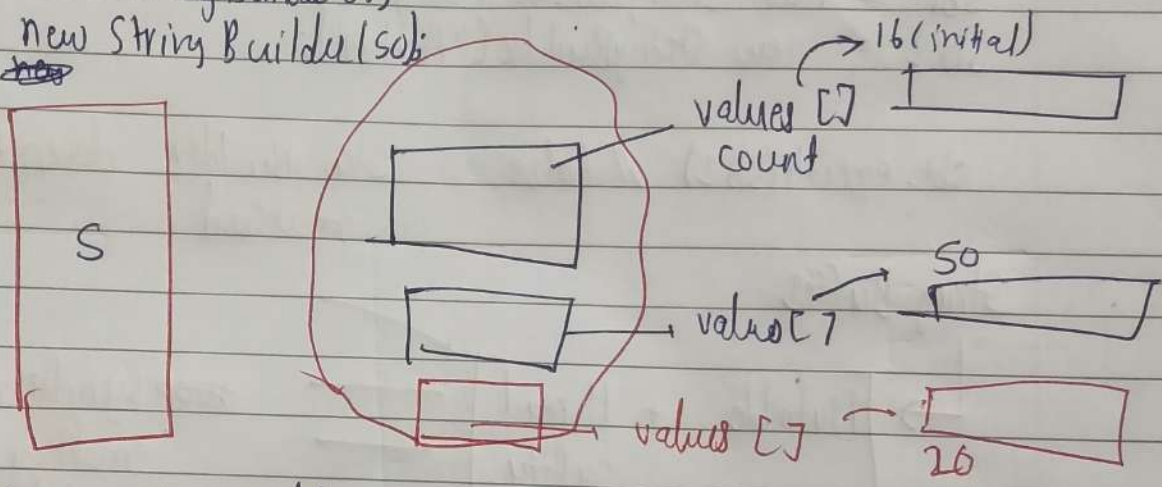
StringBuilder sb = new StringBuilder();

→ java
sb.append("is cool");
sb.append( );



StringBuilder

s = ↳ new StringBuilder();
new StringBuilder("so");



→ new StringBuilder("java");
↳ 16 + 4 = (20)

StringBuilder / StringBuffer methods

→ append()
→ insert()
→ delete()
→ replace()
→ reverse()
→ charAt()
→ setCharAt()
→ length()
→ capacity()
→ ensureCapacity()
→ trimToSize()

StringBuilder
StringBuffer
methods

```
sb = new StringBuilder();
sb.append("Hello");

String s = sb.toString();
```

transformation / comparison methods of String → don't come in StringBuilder / StringBuffer

equals()

```
Sb1 = new StringBuilder("Aditya");
Sb2 = new StringBuilder("Aditya");

sb1.equals(sb2) // false
```

StringBuilder doesn't override equals method

StringBuffer

→ [Mutable + Thread safety] — synchronize methods for thread safety

Thread 1 | Thread 2

StringBuilder sb = new StringBuilder("Hello");

Thread 1 : sb.append("A");
Thread 2 : sb.append("B");

→ HelloA
→ HelloAB

HelloB
HelloBA

Race condition

StringBuilder >> StringBuffer
fast

StringBuilder ↓ usage: 90%

StringBuffer ↓ 10% → overhead